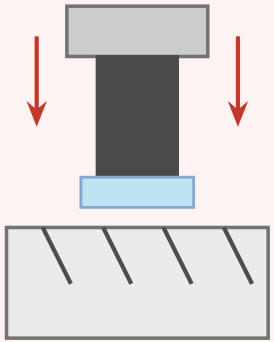
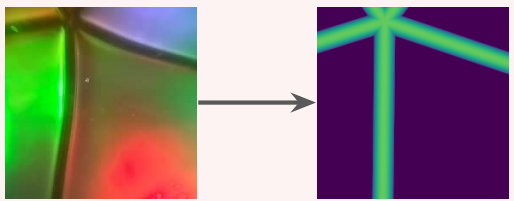
(a) Register real contacts



q^* read from the robot pose

kinematic, mounting and placement errors; redo for each session

project mesh at q^*

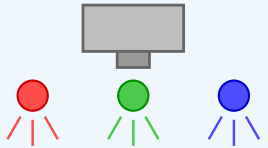


real image

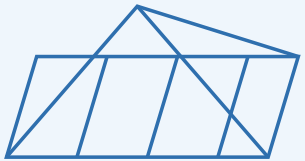
registered label

cost: one registered pose per label

(b) Simulate contacts



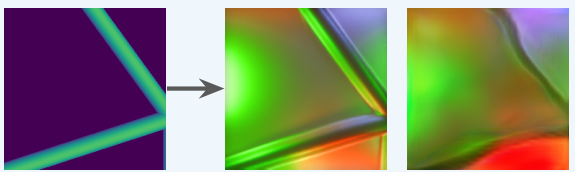
camera, LEDs, gel and contact model: calibrate per sensor



or fit a learned renderer on real image–label pairs

mesh: free geometry

render



sim depth (label)

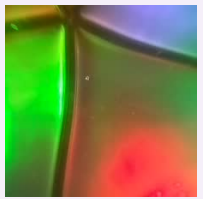
PBR

learned

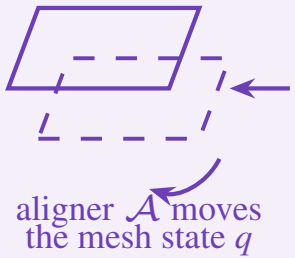
cost: calibration, or real pairs for a learned renderer; appearance gap

(c) AlignTac (ours)

learned from K registered contacts by perturbing q^*

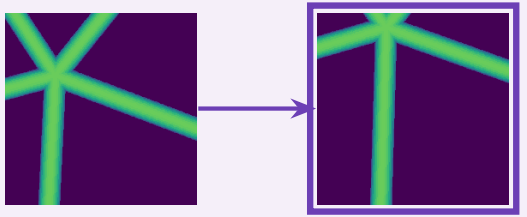


real image, fixed



aligner $\mathcal{A}$ moves the mesh state q

align



mesh at teacher pose

aligned mesh = label

cost: K registered contacts; the rest stay unlabeled